MIL-G-11725B(AT)
20 August 1985
SUPERSEDING
MIL-G-11725A(MO)
22 December 1965

MILITARY SPECIFICATION

GUN, TWIN 40 MM, SELF-PROPELLED, M42A1

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the overhaul of one type of full-tracked, armored combat vehicle, which mounts twin 40 MM guns as primary weapons.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2350

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SPECIFICATIONS
FEDERAL

VV-G-76	- Gasoline, Automotive.

MILITARY

MIL-P-514	- Plates; Identification, Transportation Data, and Blank.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-S-10379	- Suppression, Radio Interference, General Requirements for Vehicles (and Vehicular Sub-Assemblies).
MIL-S-12158	- Sealing Compound, Non-Curing, Polybutene.
MIL-G-45178	- Gun, Self-Propelled, Full-Tracked: Twin 40MM, M42 and M42A1; Preparation for Storage and Shipment of.

STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures Tactical Vehicles (Tracked and Wheeled).
MIL-STD-210	- Climatic Extremes for Military Equipment.
MIL-STD-417	- Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Vehicles.

ENGINEERING PARTS LIST

7968000	- Gun, Twin 40MM, Self-Propelled, M42A1.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First overhauled vehicle. Unless otherwise specified, the contractor shall furnish the first overhauled Gun, Twin 40mm, Self-Propelled, M42A1 which shall be subjected to a first overhaul vehicle inspection. The first overhauled vehicle offered for inspection shall be representative of the overhauled vehicles to be furnished to the Government. All subsequent M42A1's delivered to the Government shall conform to the first overhauled vehicle accepted that complies with requirements of first overhaul vehicle inspection in all their pertinent physical and performance attribures (see 4.2).

3.1.1 Workmanship and reliability. The workmanship shall be of a quality to assure that the vehicle and components thereof are free from any defects that compromise, limit, or reduce the vehicle or component capabilities herein expressed. Vehicles overhauled in accordance with this specification shall be subjected to examinations and tests to prove vehicle capabilities to perform in accordance with the following:

a. Vehicle shall require only organizational (direct support) maintenance during the first 1500 miles of normal operation, after overhaul.
b. Vehicle shall not require replacement or major repair of any automotive component during the first 3000 miles of normal operation after overhaul.
c. Replacement of track shoe assemblies shall not exceed 5 percent of each full track during the first 2000 miles of operation after overhaul.
d. The vehicle(s) mean miles between failure (MMBF) shall not be less than 385 miles during the first 3000 miles of operation. The 3000 miles of vehicles operation combat loaded or loaded with simulated load of equal weight shall consist of the following:

    1. 25 percent (%) on paved roads, either concrete or asphalt or any combination of the two.
    2. 25% on gravel and dirt roads with at least 10 percent of this distance under mud conditions.
    3. 25% on level cross-country.
    4. 25% on hilly cross-country.

e. A failure is defined as any malfunction of the end item that requires corrective action that cannot be deferred:

    1. Until the next scheduled maintenance (exclusive of lubrication services) if organizational maintenance is prescribed for correction or...
    2. For the remainder of its specified life before replacement, rebuild or salvage (as applicable) if direct or general support maintenance is prescribed for correction.

f. Corrective action is not deferrable if malfunction causes (or would cause if not corrected):

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1. An inability to commence operation, cessation of operation or degradation of performance capability of the end item below that required for the accomplishment of its prescribed mission function or...
2. A critical or catastrophic hazard to personnel or equipment.

g. Any malfunction which the driver or crew can repair will not be considered a failure provided the repair is authorized as a driver or crew function and can be accomplished within 1 hour using only controls, tools or repair parts carried with or incorporated in the end item (vehicle).

3.1.2 Maintainability. The total scheduled maintenance including driver and crew checks and services shall not exceed 75 manhours and the total unscheduled maintenance shall not exceed 35 manhours during the first 3000 miles of operation [25% hard surface road (hsr)] [25% secondary roads (sr)] [25% cross country, hilly (cch)] [25% cross country, level (ccl). The scheduled maintenance intervals shall not be less than 90 days or 750 miles whichever comes first.

3.2 Materials.

3.2.1 Qualified products. With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed on the EPL cited in 3.3, and necessary in the construction of this vehicle, the contractor shall be responsible for using only those products which are listed [by part or drawing number as Qualified Products List (QPL) items] on the QPL in the referenced specification or which have been approved for inclusion on such lists. Manufacturer's inspection records shall specifically list such component(s), name of supplier(s) (as listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL (see 4.1.4).

3.2.2 Ozone resistance. All new rubber components which are under tension or which may be flexed shall be ozone resistant to the degree required by the "C_1", test requirement of MIL-STD-417. Rubber components controlled by detailed specifications shall be ozone resistant only when required by the specification referenced on the drawing (see 4.1.4).

3.3 Overhaul. Vehicles, components, sub-assemblies, and assemblies shall be overhauled and assembled into a complete vehicle in accordance with contract, Engineering Parts List (EPL) 9968000, and overhaul technical data package. All parts requiring identification shall be identified in accordance with MIL-STD-130. Overhaul data plates shall be applied as specified (see supplement 1).

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3.3.1 Sealing.

3.3.1.1 Sealing compound. The sealing compound conforming to MIL-S-12158, shall restrict the entrance of water into the hull when the vehicle is fording streams or rivers during cross country operation and when operating in deep water (see 4.5.1.2).

3.3.1.2 Seals. Seals shall prevent the entrance, into the bearings of all foreign matter when fording shallow or deep water. All seals shall prevent the leaking of lubricants from the bearings (see 4.5.1.2 and 6.5).

3.3.1.3 Shroud seal. The space between the upper metal surface of the engine shroud and the sealing surface of the top deck grille shall prevent the recirculation of warm air (see 4.5.1.2).

3.3.1.4 Hatch seals. All hatch seals shall be free from damage, deterioration, and when installed shall prevent any leakage of water into the crew compartment (see 4.5.1.3).

3.3.2 Stowed equipment. The holding spaces provided for the On Vehicle Equipment (OVE) items shall accept and hold in place all stowed items during all conditions of vehicle operation (see 4.5.2.4).

3.3.3 Engine. The engine shall not have excessive vibration at idle or higher speeds. With engine under load, maximum governed speed shall reach 2800 revolutions per minute (rpm) but not exceed 2840 rpm. During operation of the vehicle involving all gear and speed ranges, the engine shall meet all of the functional requirements. The engine shall be free of exhaust manifold, fuel and lubricant leaks and shall be properly installed (see 4.5.1.4).

3.3.3.1 Cooling system. With vehicle operating in ambient temperature of 115 degrees Fahrenheit (°F), the cooling system shall be capable of maintaining engine oil temperatures between 180 and 225°F, and transmission oil temperature between 200°F and 285°F. Oil temperature shall be measured at point of entry to the heat exchanger (see 4.5.1.4).

3.3.4 Fuel system. During engine operation, the fuel system shall maintain fuel supply to the engine when ascending and descending 60% grades in forward and reverse drives, and when vehicle is being operated on 30% side slopes, with each side of vehicle up slope and all vehicle attitudes within the noted percentile limits (see 4.5.1.6).

3.3.4.1 Fuel tanks. Provisions shall be made to assure the internal cleanliness of the fuel tanks prior to initial fueling of vehicle. The fuel tanks shall be capable of receiving 50 gallons of fuel per minute (see 4.5.1.6).

3.3.4.2 Fuel and hydraulic lines. Provisions shall be made to assure internal cleanliness of fuel and hydraulic lines, and connections and tanks shall be free from leakage (see 4.5.1.6).

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3.3.4.3 Heater. The heater shall be capable of starting and operating satisfactorily at a temperature of minus 65°F, when burning gasoline conforming to MIL-G-3056 as a fuel for the heater. Separate supply lines, fuel pump and filter shall be used to furnish fuel to the heater from the vehicle main fuel tanks (see 4.5.1.7).

3.3.4.4 Throttle linkage. With throttle linkage attached to engine and throttle pedal fully depressed the throttle shall be in full open position. When the throttle is in full open position the throttle pedal shall be in contact with the pedal stop (see 4.5.1.8).

3.3.5 Exhaust system. All exhaust system connections and components shall be gas tight and leakproof and shall prevent the concentration of carbon monoxide in the occupied portions of the vehicle, in excess of 0.01% under any condition of operation (see 4.5.3.3).

3.3.6 Controls. All vehicle mechanical, electrical and hydraulic controls shall operate and function smoothly and effectively during all required vehicle position attitudes and operational uses (see 4.5.1.9).

3.3.7 Adjustment mechanisms. All adjustment mechanisms shall be capable of reset and shall function and operate as adjusted, maintaining the proper reset during all vehicle operations (see 4.5.1.10).

3.3.7.1 Driver's and commander's hatch. The driver's and commander's hatch shall break away and slide through its full range of travel without undue effort or indications of binding or interference. Hatch shall lock securely in the open and closed position throughout all vehicle operating conditions, and in all vehicle running and parked attitudes (see 4.5.1.11).

3.3.7.2 Front door. Front door shall be properly aligned and operate through its full range of travel without undue effort or indications of binding, and shall lock securely in the open and closed position. Seal shall be free of damage and deterioration, and when installed shall prevent entrance of water into vehicle (see 4.5.1.11).

3.3.7.3 Driver's seat. All sliding parts and operating contact surfaces shall be free of paint. The driver's seat shall operate through its horizontal and vertical adjustments without undue effort or indications of binding. The driver's seat shall move forward, backward, vertically, or dump when the appropriate levers are activated (see 4.5.1.12).

3.3.7.4 Front and rear drain valves. Drain valves and controls shall operate freely, drain valves shall open and close without indications of binding or sticking. Drain valve controls must be capable of locking drain valves in the open position (see 4.5.1.13).

3.3.7.5 Air cleaner. All components of the air cleaner system as installed in the vehicle shall be in good condition, and correctly installed and shall prevent the entrance of all foreign matter (see 4.5.1.14).

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3.3.8 Vision devices and receptacles. Receptacles incorporated in the hull, including driver's hatch, shall accept applicable vision devices (normally installed in the field), without binding or interference. Locking devices shall function properly and secure vision device installed in the receptacle. When installed, the vision devices in the receptacle shall prevent entrance of water into the vehicle (see 4.5.1.15).

3.3.9 Fire extinguisher system. Seals shall not be broken. State of charge of each cylinder shall be properly recorded. Release mechanism shall actuate the system without undue effort on the release handle. System shall discharge properly and shall be free of CO_2 leaks. All items and components of the fire extinguisher system shall be correctly installed (see 4.5.2.5).

3.3.10 Ammunition racks. The hull ammunition stowage racks shall be free of all defects and conditions adversely affecting appearance and proper stowage and retention of ammunition (see 4.5.1.16).

3.3.11 Turret.

3.3.11.1 Gunner's seats. All sliding parts and operating contact surfaces shall be free of paint. The seat shall be free of binding and interference throughout its operating range. Locking devices shall function properly and shall hold seat in the desired position throughout all operating conditions. CAUTION: Operator shall be sitting in seat when it is raised or lowered (see 4.5.1.17).

3.3.11.2 Loaders seats. With the seat properly installed and not in use, the lifting spring shall be capable of moving and retaining the seat in a vertical position (see 4.5.1.17).

3.3.11.3 Turret ammunition racks and boxes. The turret ammunition racks and boxes shall be free of all defects and conditions adversely affecting appearance or proper stowage and retention of ammunition (see 4.5.1.18).

3.3.11.4 Cartridge case chutes and deflectors. The chute and deflectors shall be free of all defects and conditions adversely affecting appearance or proper ejection of cartridge case (see 4.5.1.18).

3.3.11.5 Turret components. Turret components shall be correctly installed (see 4.5.1.19).

3.3.11.6 Foot control firing mechanism. The foot control firing mechanism shall be correctly installed. When depressing the foot pedal there shall be no binding or excessive free travel in linkage (see 4.5.1.19).

3.3.11.7 Elevating and traversing locks. The elevating and traversing locks shall engage and disengage without interference and shall lock the gun and mount securely (see 4.5.1.19).

3.3.12 Welding repairs. Welding repairs of any type or class shall be made only when, and to the extent, specifically authorized by the procuring activity (see 4.5.1.20).

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3.3.13 Electrical system. Each circuit shall maintain continuity from end to end without any shorts from pin to other pins and from each pin to the connector shell (see 4.5.1.21).

3.3.13.1 Generator voltage. The engine shall be operated at 1000 rpm for 30 minutes before setting voltage regulator. With the vehicle generator operating at a nominal temperature, it shall deliver between 27.8 and 28.3 volts (V) direct current (dc) at regulator under all conditions of vehicle operation (see 4.5.1.21).

3.3.13.2 Slip ring (hull to turret) and external communication circuits. The slip ring and external communication circuits shall maintain continuity without evidence of shorts. Each circuit shall maintain continuity from end to end, without any shorts from one pin to the other and from each pin to the connector shell (4.5.1.21).

3.3.13.3 Main gun firing circuits. The main gun firing solenoid shall be capable of providing the energy necessary to detonate main round (see 4.5.1.21).

3.3.13.4 Lights. The internal and external lights as specified on applicable drawings shall operate as specified through all vehicle operating conditions. All electrical contacts and connections shall maintain positive contact under all vehicle operating conditions (see 4.5.1.21).

3.3.13.5 Auxiliary generator. The auxiliary generator consisting of an engine, generator, and controls shall operate as starter generator combination. With generator operating in nominal ambient temperature, it shall deliver between 27.8 or 28.3 V dc at regulator under all condtions of vehicle operation (see 4.5.1.21).

3.3.13.6 Personnel heaters and bilge pump. Personnel heaters and bilge pump shall function properly in starting, operating and stopping, under all conditions of vehicle operation (see 4.5.1.21).

3.3.13.7 Infrared power pack high voltage supply. With the generator supplying specified voltage, the infrared power pack with an actual load or simulated load, output voltage at the connector shall be a minimum of 16 000 V, without any evidence of high voltage arcing (see 4.5.1.22).

3.4 Vehicle break-in run.

3.4.1 Preparation. Prior to break-in run, each vehicle shall be completely assembled and properly lubricated as specified by lubrication order. Fuel tanks shall be filled to the level necessary to complete the break-in run and the required performance testing of the vehicle. Fuel used shall conform to VV-G-76 or MIL-G-3056. Engine governor shall be set to provide for operation of the engine between 2800 and 2900 rpm at full load, full throttle. The no-load operation shall be limited to 3000 rpm. The vehicle shall be examined for fuel and lubricant leaks. All other necessary adjustments and safety checks shall be completed (see 4.3).

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3.4.2 Oil pressure and temperature. Prior to the start of the break-in run, the engine shall be operated at 800 to 1500 rpm until lubricating oil is at specified operating pressure and temperature. If at any time during the break-in run the engine has been stopped for at least 30 minutes, the engine shall again be operated, as above, before continuing break-in run. Proper oil pressure and temperature shall be maintained during operation on level ground, 60% grades and 30% side slopes, and all vehicle attitudes within the noted percentile limits (see 4.3).

3.4.3 Operation and distance. Prior to tests, each vehicle shall be given a break-in run for the distances specified in table I on smooth, level, hard-surfaced roads or test track. The vehicle shall be operated over the undulating section of road (see figure 1) during "a" and "b" divisions of the break-in run, twice in each direction (see 4.3).

TABLE I. Speeds and distances for break-in run.

Division of run	Speed mph	Distance miles	Test conditions
a	0 to 10	5	Track (road)
b	11 to 20	7 1/2	Track (road)
c	21 to Max. Governed	7 1/2	Track (road

3.4.4 Reverse operation. After each division of the break-in run, vehicle shall be stopped, the engine allowed to idle for not less than 2 minutes, and the vehicle driven in reverse for a distance of not less than 50 feet at engine speed of not more than 1500 rpm (see 4.3).

3.4.5 Condition after run. After completion of the break-in run, the engine shall run smoothly when operated between 750 and 900 rpm. Prior to vehicle being submitted to the Government for acceptance, all suspension wheel bearings shall be checked to determine if adjustments are correct and readjusted as required. There shall be no defects indicative of damaged components, or maladjustments that may cause faulty vehicle operation (see 4.3.1).

3.5 Performance. A complete vehicle, combat loaded (see 3.3.2) and serviced shall perform as specified herein. Ammunition may be simulated in size and weight. Vehicle, serviced (see 3.4.1) and equipped for existing climatic conditions, shall operate as specified without special equipment. Vehicle shall be capable of operating on 60 percent longitudinal slopes, headed up and headed down slopes, and on right and left 30% side slopes, with each side of vehicle up slope (see 4.5.1.23).

3.5.1 Environmental. The vehicle shall operate in ambient temperature of plus 115°F to minus 25°F, and with special equipment installed at temperature range of minus 25°F to minus 65°F. The complete vehicle shall withstand storage at minus 80°F and plus 160°F without deterioration that may cause failure of any component part of the vehicle (see 4.5.3.5 and 4.5.3.6).

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3.5.2 Power package and power train. When power package is installed in accordance with overhaul technical data package, it shall function throughout all gear and speed ranges without loss of lubricants, grabbing, and chattering or slippage when controls are applied to control, stop, and hold vehicle (see 4.5.1.23).

3.5.3 Speeds.

3.5.3.1 Level road speeds. The vehicle shall operate at sustained speeds of 25 mph and shall be capable of attaining not less than 40 mph, and at 2.5 mph on smooth, level relatively hard-surfaced roads without damage to power plant and power train. When traveling between 15 and 30 mph the vehicle drift shall not exceed 3 feet in 100 feet (see 4.5.1.24).

3.5.3.2 Grade speeds. The vehicle shall operate at a sustained speed of 10 mph while ascending a 10% grade, and at 20 mph while ascending a 3% grade (see 4.5.3.7).

3.5.4 Acceleration. The vehicle shall accelerate from a standing start on level ground to 20 mph in not more than 12 seconds (see 4.5.1.25).

3.5.5 Engine starting on grades and slopes. When standing on a 60% grade, for not less than 2 minutes, with engine operating under no load between 700 and 750 rpm, the engine shall be stopped for not less than 2 minutes. The engine shall restart in not more than 1 minute when headed up and headed down grade. Similar engine stopping and starting shall be accomplished on 30% right and left side slopes, and any combinations of vehicle attitude within noted percentile limits (see 4.5.1.26).

3.5.6 Braking and drift.

3.5.6.1 Stopping. Vehicle traveling at 20 mph shall stop within 45 feet from point of brake application; drift not to exceed 4 feet when stopping (see 4.5.1.27).

3.5.6.2 Holding. With vehicle combat loaded, (see 3.1.1.1) or a simulated load of equal weight and standing on a 60% grade with service brakes applied, the vehicle shall be held stationary when headed up and headed down grade. With parking brake engaged and all other holding devices inoperative, the vehicle shall be held stationary when headed up and headed down grade (see 4.5.1.28).

3.5.7 Turning. The vehicle shall pivot 360 degrees to the right and left in neutral steer within a circle 35 feet in diameter (see 4.5.1.29).

3.5.8 Fording.

3.5.8.1 Shallow water. Vehicle shall ford a level, hard-bottom body of clear water 40 inches in depth without special equipment. With vehicle standing in water 40 inches in depth for 30 minutes, the accumulation of water shall be not more than 1 inch on the crew compartment hull floor measured forward of the bulkhead (see 4.5.1.30).

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3.5.8.1.1 Engine operation. With the vehicle engine compartment flooded with 40 inches of clear water (from ground level) and engine operating at 1000 rpms for at least 15 minutes it shall be stopped and then restarted within a time frame of 3 minutes. Engine and all accessories shall function satisfactorily during and after fording operations (see 4.5.1.30).

3.5.8.2 Lubrication contamination. All accessories shall function satisfactorily during and after fording operations. After fording operations, the water contamination content of the transmission, engine, final drives, and suspension system lubricants shall be not more than 2 percent by volume (see 4.5.2.3).

3.5.8.3 Bilge pumps. The bilge pumps, installed in accordance with applicable drawings, shall remove water from the vehicle at a minimum rate of 50 gallons per minute (gpm) each, with a discharge head of 4 1/2 feet and with pump motor operating on 27.5 V impressed at the motor (see 4.5.1.30).

3.5.9 Trench crossing. The vehicle shall be capable of crossing trenches 36 inches or more in depth and 72 inches in width without stalling or damage to the vehicle including but not limited to suspension, gun tube, fenders and hull (see 4.5.3.8).

3.5.10 Vertical obstacles. The vehicle shall cross over vertical obstacles 28 inches in height without stalling or damage to suspension and hull floor (see 4.5.3.9).

3.5.11 Climbing. Vehicle shall ascend a prepared 60% longitudinal grade in forward and reverse drive from a stand-still position, and 30% right and left side slopes, or any vehicle attitude within noted percentile limits without stalling or damage to power train and power plant (see 4.5.3.10).

3.5.12 Vehicle turret and gun control.

3.5.12.1 Turret and gun control system.

3.5.12.1.1 Manual traverse effort. With the gun horizontal and mount or carriage level, the traversing mechanism shall operate smoothly and with uniform effort. The torque required at hand crank shaft to traverse the mount or carriage in either direction shall not exceed 90 inch pounds (see 4.5.1.31.1).

3.5.12.1.2 Traverse mechanism backlash. Backlash in the traversing mechanism shall be as follows: With the gun approximately horizontal and mount or carriage level, the backlash shall not exceed 1/2 mil at the gun muzzle, corresponding to 2 degrees and 5 minutes at the oil gear shaft (see 4.5.1.31.2).

3.5.12.1.3 Power traverse. With 27.5 V applied at slip ring the control system M16A1E1 shall operate the turret clockwise and counterclockwise through 360 degrees traverse without excessive surge or erratic movement. With control handle in full slew position, the traverse speed shall be not less than one complete revolution of turret in 9 seconds. Rate of movement shall be the same in both directions (see 4.5.1.31.3).

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3.5.12.1.4 Elevation and traverse system stability. When turning rate motor toggle switch on drive controller to the ON position the rate motor should start immediately. With rate motor running and drive controller grip assemblies in neutral position, the amount of creep or gun mount movement shall not exceed 675 mils per minute in azimuth or 3 degrees per minute in elevation (see 4.5.1.31.4).

3.5.12.1.5 Manual elevation effort. The torque required at the hand crank shaft, to elevate or depress the gun, shall not exceed 90 inch-pounds between 5 degree elevation and 80 degree elevation. Torque shall not exceed 125 inch-pounds at points below 5 degree elevation and above 80 degree elevation. Elevating mechanism shall operate smoothly through full range of elevation (see 4.5.1.31.5).

3.5.12.1.6 Elevation mechanism backlash. Backlash in the elevating mechanism shall be as follows: With the gun approximately horizontal and mount or carriage level, the backlash shall not exceed 1/2 mil at the gun muzzle, corresponding to 3 degrees and 17 minutes at the oil gear shaft (see 4.5.1.31.6).

3.5.12.1.7 Manual elevation limit stops. The elevation limit stops shall be set to provide vertical movement of the dual gun from -5° to + 87° (see 4.5.1.31.7).

3.5.12.1.8 Power elevation. Elevate and depress the guns by rotating the grip assemblies forward for depression or backward for elevation. Gun movement shall be smooth without erratic action. The hydraulic limit stop assembly in elevation oil gear shall limit gun movement to minus 3° and plus 85°. The guns shall elevate from a position of 20° to a position of 80° in 2.4 seconds and depress from a position of 60° to a position of 0° in 2.4 seconds (see 4.5.1.31.8).

3.5.13 Gun fire control system.

3.5.13.1 Synchronization. The sight computer M38 leveling pads shall be parallel to the gun mount carriage leveling pads within ± 2 mils from maximum depression to 45 degree elevation. From elevations of 45 degree to maximum elevation, leveling pads of sight computer M38 and leveling pads of gun mount carriage shall be parallel within ± 4 mils (see 4.5.1.32.1).

3.5.13.2 Boresighting. The boresighting lines of sight of the sighting instruments shall be in line with a line of sight through the axis of the gun bores upon a common aiming point not less than 1000 yards distant. The synchronization and boresighting of the gun and computor shall insure the accuracy of hitting target (see 4.5.1.32.2 and 4.5.1.32.3).

3.5.14 Radio suppression. Vehicle shall be radio interference suppressed in accordance with tactical vehicle requirements of MIL-S-10379 (see 4.5.3.11).

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3.6 Painting. Unless otherwise specified, the exterior of vehicle, components, assemblies, sub-assemblies, parts that require painting, and interior of vehicle shall be prepared and painted in accordance with MIL-STD-193. Vehicle exterior paint and exterior non-skid paint shall match color chip X24087 of FED-STD-595.

3.7 Marking. Marking of vehicle shall be in accordance with MIL-STD-642.

3.8 Data plates. All data plates shall be in accordance with MIL-P-514.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Parts and components. Parts, components and assemblies shall be inspected in accordance with the drawings listed in overhaul technical data package.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to determine quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Qualified product. The contractor's inspection records shall be checked to determine contractor conformance to 3.2.1 and 3.2.2.

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4.2 First overhaul vehicle inspection.

4.2.1 Special process inspection. A special process inspection will be conducted by the Government representatives during overhaul of the first vehicle (see 3.1) to evaluate conformance of materials and workmanship with drawing and specification requirements.

4.2.1.1 Review and evaluation. Review and evaluation of records for processing procedure, inspection systems and plan of quality control, will be conducted on the following: material, welding, surface hardening, heat treating, protective finishes, etc.

4.2.1.1.1 Hull inspection. The hull shall be inspected prior to covering the weldments with paint or other pertinent components or assemblies.

4.2.1.2 Complete vehicle inspection. The first vehicle when complete, shall be inspected by the Government subsequent to contractor's inspection at the contractor's plant to determine conformance to contract and specification requirements (see 3.1).

4.2.1.2.1 Corrections. Required corrections found to be necessary as a result of complete vehicle inspection by the Government, as stipulated in paragraphs 3.1 and 4.1.3, shall be made by the contractor on the submitted first vehicle. The same vehicle with all required corrections incorporated shall be resubmitted for Government inspection.

4.2.1.3 Overhaul production vehicle test. After completion of the first overhauled vehicle (see 3.1) inspection, a second vehicle shall be selected from the first month's overhaul production, or one of the first 10 and shall be subjected to examination (see 4.4.2) and tests specified in 4.5 (a) and (b). Subsequent to examination and tests, the selected vehicle shall be subjected to the 3000 mile test specified in table II at a site approved by the Government, test will require 90 days.

TABLE II. 3000 Mile test combat loaded (Government Proving Ground).

Course	Mileage and speeds
*Hard surface roads	750 miles at varying speeds up to maximum
Gravel and dirt roads	750 miles at varying speeds up to maximum
Level cross country	750 miles at varying speeds up to maximum
*Hilly cross country	750 miles at varying speeds up to maximum

*NOTE: Hard surface road and hilly cross country mileage includes 20 miles with towed load.

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4.2.2 Test failure. Failure of the vehicle as the result of any deficiency of a workmanship or materials nature during or as a result of the 3000 mile test, shall be cause for rejection of the vehicle. Further, the Government will refuse to continue acceptance of overhauled vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the deficiency.

4.3 Break-in run (track). To determine conformance to 3.4.1, 3.4.2, and 3.4.3 each vehicle shall be lubricated and serviced prior to the break-in run. The run shall be on smooth, level, hard-surface roads, and for the distance specified in table I. The vehicle shall be operated in reverse after each division specified in 3.4.4. The undulating portion of the test shall be accomplished twice in each direction at speeds between 5 and 10 mph during division "a" and between 11 and 14.5 mph during division "b".

4.3.1 Condition after run. To determine conformance to 3.4.5, after completion of the break-in run, the vehicle shall be inspected to determine that there is no damage or that any operating elements do not malfunction, and all adjustments are maintained as required.

4.4 Inspection provisions.

4.4.1 Lot size. An inspection lot shall consist of all vehicles of one type from one day's production, submitted at one time for acceptance inspection.

4.4.1.1 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling may be accomplished by the contractor, the contractor shall 100% inspect the first twenty vehicles to establish a process average (see 6.4) to allow normal sampling in accordance with MIL-STD-105.

4.4.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examinations shall be performed against the classification of defects and with the acceptable quality levels shown in table III. Functional defects found in properly installed Government furnished equipment will be classified as special and shall not be used as a basis for rejection of the vehicle. The following constitutes a part of the classification of defects (see 6.5):

a. Any drip from the inside to the outside of a self-contained reservoir constitutes major defect when the vehicle has been standing idle and the components are at ambient temperature.

b. Any seep that occurs from the inside to the outside of a static fit (metal to metal, or gasket combination) shall constitute a major defect.

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TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None defined.	
Major	AQL 25 Defects/100 Units	
101	Engine, malfunction, leaks (see 6.6c), mounting improper, damage	Visual and functional
102	Engine governors, improper setting, malfunction.	Functional
103	Power train, malfunction, mounting improper, damage leaks (see 6.6c).	Visual and functional
104#	Fuel system components malfunction, damage leaks (see 6.6c).	Visual and
105	Cooling system components (engine and transmission) malfunction, damage leaks (see 6.6b).	Visual and functional
106	Exhaust system components damage, leaks.	Visual
107	Electrical system components (including wiring and instrument malfunction, damage.	Visual and functional
108	Suspension system components, damage, tire or track pad base separation, improper clearance or alignment, welding defects leaks (see 6.6c), malfunction.	Visual
109	Hull, structure and welding defects (metal shrink or crater cracks excluded).	Visual
110	Controls and adjusting mechanism malfunction, improper clearance.	Visual and functional
111	Doors, fenders, grilles, panels, ducts, stowage brackets, boxes, shocks and stops, malfunction: torsion spring and hold-open devices.	Visual and functional
112	Drain valves, valve seating improper.	Visual and functional
113	Cushions, seats, rests and crash pads, seat control malfunction, damage.	Visual and functional
114	Turret traverse lock, locking malfunction, damage.	Visual and functional
115	Traversing and elevating mechanism, clearance improper, component damage.	Visual and functional
116	Main armament, component damage, improper assembly or trunion cap bolt torque.	Visual and
117	Weapon control system (includes panels, solenoids, switches, wiring and other components) component damage, defective wiring.	Visual and functional

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TABLE III. Classification of defects - Continued

Category	Defect	Method of examination
118	Miscellaneous items or accessories malfunction, damage, missing.	Visual and functional
119	Fire extinguisher system components, paint-protective and antiskid coatings, lube fittings and lubrication. Decals, markings, data and instruction plates, seals missing or defective.	Visual
Minor	AQL 150 Defects/100 Units	
201	Engine, improper component assembly or installation.	Visual and functional
202	Engine governors, missing or defective seals.	Visual and functional
203	Power train, improper component assembly or installation.	Visual and functional
204#	Fuel system components improper assembly, installation or clearance.	Visual and functional
205	Cooling system components (engine and transmission), improper assembly or installation.	Visual and functional
206	Exhaust system components, improper assembly installation or clearance, leaks.	Visual
207##	Electrical system components (includes wiring and instruments) improper assembly installation, circuit identification, clearance or protection, illumination of reticle.	Visual and functional
208	Suspension system components, improper assembly installation, bolt or nut torque and track tension.	Visual
209	Hull, controls and adjusting improper assembly or installation.	Visual and functional
210	Doors, grilles, fenders, panels ducts, stowage brackets, boxes, shocks and stops, improper installation, assembly installation or fits.	Visual and functional
211	Drain valves, improper assembly installation or control adjustment.	Visual and functional
212	Cushions, seats, rests and crash pads, improper installation, fit adhesions or damage.	Visual and functional
213	Turret traverse lock, improper assembly, installation or adjustment.	Visual and functional
214	Traversing and elevating mechanism main armament improper installation or assembly.	Visual and functional
215	Weapon control system (includes panels, solenoids, switches wiring and other components), improper clearance, assembly installation, circuit identification, protection or markings.	Visual and functional

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TABLE III. Classification of defects - Continued.

Category	Defect	Method of examination
216	Miscellaneous items or accessories improper assembly or installation.	Visual and functional
217	Fire extinguisher system components data card incomplete, improper assembly or installation.	Visual
218	Paint-protective and anti-skid coatings, application coverage and materials improper.	Visual
219	Lube fittings and lubrication fittings missing, defective or improperly installed.	
220	Decals, markings, data and instruction plates, incomplete data missing, improper location or site illegible.	Visual

Fuel leaks constitute a special defect and will result in total inspection of the lot for this deficiency.

Filament failures are not to be classified as a major or minor defect.

4.4.2.1 Unclassified defects. All component defects having no bearing on function, safety, interchangeability or life, but which are considered departures from good workmanship, will be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots, or in ten lots or more within a thirty-day period, will be added to the minor classification with no decrease to AQL's.

4.4.2.2 Recurring major deficiencies. A major deficiency (see 6.6) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

4.4.2.3 Recurring minor deficiencies. A minor deficiency (see 6.6) is recurring if it occurs more than twice in the same sample or when the defects occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. Correction of all deficiencies shall be accomplished prior to acceptance.

4.5 Classification of tests.

a. Acceptance tests, 4.5.1.1 through 4.5.1.32.1.
b. Control tests, 4.5.2 through 4.5.2.5.
c. Inspection comparison tests (ICT), 4.5.3 through 4.5.3.12.

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4.5.1 Tests.

4.5.1.1 Acceptance tests. To determine conformance to 3.5, after break-in run (see 3.4), each vehicle shall be operated for a distance of 8 miles or more and shall be inspected as specified in 4.4.2. These acceptance tests shall include all acceptance tests specified at the place of manufacture in table IV. Except for equipment and component installation and servicing performed by the contractor as a final processing of vehicles for acceptance, vehicles shall be completely assembled, all adjustments completed, and serviced with fuel, lubrication and fluid as specified, but need not be combat loaded except for holding test (see 4.5.1.28). After the tests, the vehicle shall be examined for evidence of fuel or lubricant leakage, or other deficiencies.

4.5.1.1.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government shall stop acceptance until evidence has been provided by the contractor that corrective action has been taken.

TABLE IV. Location for tests.

Title	Paragraph Number	Place of Manufacture	Government Proving Ground
Acceptance tests	4.5.1.1	X	X
Sealer and seals test	4.5.1.2	X	X
Hatch seal test	4.5.1.3	X	X
Engine test	4.5.1.4	X	X
Fuel system slope check	4.5.1.5	X	X
Fuel and hydraulic line check	4.5.1.6	X	X
Heater check	4.5.1.7	X	X
Throttle linkage check	4.5.1.8	X	X
Controls check	4.5.1.9	X	X
Adjustment mechanism check	4.5.1.10	X	X
Driver's and Commander's hatch and excape door check	4.5.1.11	X	X
Driver's seat check	4.5.1.12	X	X
Front and rear drain valve test	4.5.1.13	X	X
Air cleaner check	4.5.1.14	X	X
Vision devices and receptacle check	4.5.1.15	X	X
Ammunition racks check	4.5.1.16	X	X
Gunner's and loaders seat check	4.5.1.17	X	X
Turret ammunition racks, boxes, cartridge chutes, and deflectors check	4.5.1.18	X	X
Turret components, foot control firing mechanism, elevating and traversing locks check	4.5.1.19	X	X

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TABLE IV. Location for tests - Continued.

Title	Paragraph Number	Place of Manufacture	Government Proving Ground
Welding repairs and burning checks	4.5.1.20	X	X
Electrical system check	4.5.1.21	X	X
Infrared power pack high voltage supply check	4.5.1.22	X	X
Power package and power train test	4.5.1.23	X	X
Level road speed test	4.5.1.24	X	X
Acceleration test	4.5.1.25	X	X
Engine starting on grades and slope tests	4.5.1.26	X	X
Stopping test	4.5.1.27	X	X
Holding test	4.5.1.28	X	X
Turning test	4.5.1.29	X	X
Fording test	4.5.1.30	X	X
Turret and gun control system test	4.5.1.31	X	X
Manual traverse effort	4.5.1.31.1	X	X
Traversing mechanism backlash	4.5.1.31.2	X	X
Power traverse	4.5.1.31.3	X	X
Elevation and traverse system stability	4.5.1.31.4	X	X
Manual elevation effort	4.5.1.31.5	X	X
Elevation mechanism backlash	4.5.1.31.6	X	X
Manual elevation limit stops	4.5.1.31.7	X	X
Power elevation	4.5.1.31.8	X	X
Synchronization	4.5.1.32.1	X	X
Control tests	4.5.2	X	X
Contamination test	4.5.2.3	X	X
Stowed equipment test	4.5.2.4	X	X
Fire extinguisher test	4.5.2.5	X	X
Inspection comparison tests	4.5.3		X
Fuel system test	4.5.3.2		X
Exhaust system test	4.5.3.3		X
Communications equipment test	4.5.3.4		X
Sub-zero temperature tests	4.5.3.5		X
High temperature tests	4.5.3.6		X
Speed on grades test	4.5.3.7		X
Trench crossing tests	4.5.3.8		X
Vertical obstacle test	4.5.3.9		X
Climbing test	4.5.3.10		X
Radio suppression test	4.5.3.11		X
Carbon dioxide distribution test	4.5.3.12		

4.5.1.2 Sealing and seals check. To determine conformance to 3.3.1.1, 3.3.1.2, and 3.3.1.3, prior to and after fording operations the vehicle shall be checked for the amount of water in vehicle and the leaking of lubricants.

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4.5.1.3 Hatch seal test. To determine conformance to 3.3.1.4, the driver's, loader's, and commander's hatches shall be closed and locked, and a spray of water directed on each hatch for a period of 3 minutes. All hatches shall be checked for leakage.

4.5.1.4 Engine test. To determine conformance to 3.3.3 and 3.3.3.1, the vehicle, shall be operated as specified and observed for performance requirements, and the cooling system checked for maintaining specified temperatures at local ambient air temperature.

4.5.1.5 Fuel system slope check. To determine conformance to 3.3.4, the vehicle shall be operated on grades and slopes and observed for functional requirements.

4.5.1.6 Fuel and hydraulic line check. To determine conformance to 3.3.4.1 and 3.3.4.2, the fuel and hydraulic system shall be pressurized with dry air at a pressure of 3 to 60 psi, prior to engine installation, and checked for leaks at the fittings and fuel tank seals, by applying soapy water or equivalent.

4.5.1.7 Heater check. To determine conformance to 3.3.4.3, with the heater pump operating as specified the heater shall be checked for functional requirements.

4.5.1.8 Throttle linkage check. To determine conformance to 3.3.4.4, the throttle pedal shall be operated to full open throttle, and check the engine throttle rod adjustment to insure that full open throttle position has been reached.

4.5.1.9 Controls check. To determine conformance to 3.3.6, all controls shall be operated and checked for functional requirements.

4.5.1.10 Adjustment mechanism check. To determine conformance to 3.3.7, all adjustment mechanisms requiring adjustment shall be checked for proper adjustment.

4.5.1.11 Driver's and commanders' hatch and escape door check. To determine conformance to 3.3.7.1 and 3.3.7.2, the driver's and commander's hatch and escape door shall be closed, opened, then closed, and checked for binding, interference, and locking operation.

4.5.1.12 Driver's seat check. To determine conformance to 3.3.7.3, the driver's seat shall be operated through its complete range of operation and checked for functional requirements.

4.5.1.13 Front and rear drain valve test. To determine conformance to 3.3.7.4, the drain valves shall be operated, as specified, and checked for functional requirements in opened, closed, and locked positions.

4.5.1.14 Air cleaner check. To determine conformance to 3.3.7.5, the air cleaner shall be checked for proper installation and functional requirements.

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4.5.1.15 Vision devices and receptacles check. To determine conformance to 3.3.8, all receptacles shall be checked for acceptability of vision devices, and for leaks.

4.5.1.16 Ammunition racks check. To determine conformance to 3.3.10, all ammunition racks shall be checked for the acceptability and retention of ammunition.

4.5.1.17 Gunner's and loader's seat checks. To determine conformance to 3.3.11.1 and 3.3.11.2, the seats shall be operated throughout all ranges of operation and checked for functional requirements.

4.5.1.18 Turret ammunition rack, boxes, cartridge case chutes, and deflectors check. To determine conformance to 3.3.11.3 and 3.3.11.4, the racks, boxes, cartridge case chutes, and deflectors shall be checked for proper installation and damage.

4.5.1.19 Turret components, foot control firing mechanism, elevating and traversing locks check. To determine conformance to 3.3.11.5 through 3.3.11.7, all items shall be checked for proper installation, conditions, damage and functional requirements.

4.5.1.20 Welding repairs. To determine conformance to 3.3.12, all welding repairs shall be checked for meeting approved required repairs and removal of slag.

4.5.1.21 Electrical system check. To determine conformance to 3.3.13 through 3.3.13.6, the electrical system shall be checked using calibrated master voltage meter.

4.5.1.22 Infrared power pack high voltage supply check. To determine conformance to 3.3.13.7, with the master battery control and infrared power switches turned on and generator supplying specified voltage, the voltage shall be measured at the loose harness plug using an 80 megaohm load.

4.5.1.23 Power package and power train test. To determine conformance to 3.5 an 3.5.2, the vehicle shall be operated throughout all gear and speed ranges, the power plant, power train, final drives, tracks, and suspension system, and applicable controls shall be checked for functional requirements.

4.5.1.24 Level road speed test. To determine conformance to 3.5.3.1, the vehicle shall be operated at maximum speed for not less than 10 minutes and at minimum speed for not less than 5 minutes and checked for function of components. Drift shall be not more than that specified.

4.5.1.25 Acceleration test. To determine conformance to 3.5.4, the vehicle shall be operated as specified, the distance and time checked and recorded.

4.5.1.26 Engine starting on grades and slopes test. To determine conformance to 3.4.2 and 3.5.5, the engine shall start and oil pressure temperature shall be maintained when operating on specified grades and slopes. The 60% grade test may be conducted in conjunction with brake holding test.

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4.5.1.27 Stopping test. To determine conformance to 3.5.6.1, the vehicle shall be operated on dry level hard-surfaced road without loose material, and the results of 3 stopping tests shall be averaged to make this determination.

4.5.1.28 Holding test. To determine conformance to 3.5.6.2, a simulated combat load shall be placed on the vehicle in a position that will not restrict the flow of engine air (intake or exhaust). The vehicle shall be driven up a 60% grade and the service brakes and then the parking brakes checked for conformance. The engine starting and fuel system tests on a 60% slope may be conducted in conjunction with this test.

4.5.1.29 Turning test. To determine conformance to 3.5.7, vehicle shall be operated and turned to right and to the left in full 360 degree pivot turn in neutral steer and checked for functional requirements.

4.5.1.30 Fording test. To determine conformance to 3.5.8.1, 3.5.8.1.1, 3.5.8.2, and 3.5.8.3, the vehicle shall be driven into water of specified depth without special equipment. During or after test period, water accumulation on the crew compartment hull floor, shall be not more than specified depth. The engine shall idle at 1000 rpm for 15 minutes, be stopped, and restarted in not more than 3 minutes with all accessories functioning satisfactorily during and after fording.

4.5.1.31 Turret and gun control system test.

4.5.1.31.1 Manual traverse effort. To determine conformance to 3.5.12.1.1, attach a spring scale to hand crank and traverse mount through full range of 360° (6400 mils), in each direction, and observe operation. Effort shall not exceed specified requirements.

4.5.1.31.2 Traversing mechanism backlash. To determine conformance to 3.5.12.1.2, the traversing mechanism shall be checked as follows:

a. If a backlash gage is available, check backlash as described in (1) through (4) below:

   1. Install scale component of backlash gage on bearing adapter.
   2. Install vernier component of backlash gage on output drive coupling and adjust knurled knob until gage reads zero.
   3. Twist coupling one way until gun begins to traverse; stop, and record reading on gage.
   4. Twist coupling in opposite direction until slack is taken up without moving the gun barrel; stop, and record the reading on the gage. A boresight or suitable telescope clamped to the gun and sighted on a distant object should be utilized to detect motion of gun barrel. Backlash shall not exceed 2° and 5 minutes.

b. If a backlash gage is not available, check backlash by first placing gun approximately horizontal, clamp a suitable collimator or telescope on breech casing and proceed as described in (1) through (5) below:

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1. Attach a wire to azimuth oil gear shaft to serve as a pointer.
2. Select a distant object, traverse gun either to right or left, and sight telescope so that the selected object appears at intersecting lines on the reticle. Approach object gradually and do not overpass.
3. Without moving the gun, scribe a line on the 4 1/2 inch flange diameter of the bearing adapter to conicide with the wire pointer.
4. Traverse gun in same original direction about one-half turn of the traversing hand crank.
5. Reverse the direction of rotation until intersection of telescope reticle lines fall on original reference point but do not overpass. Position of wire pointer must coincide with scribed lines on bearing adapter within three thirty-seconds of an inch.

4.5.1.31.3 Power traverse. To determine conformance to 3.5.12.1.3, traverse gun mount slowly by moving the drive controller grip assemblies in a horizontal plane; with right grip assembly forward, the gun mount will traverse counterclockwise; with left grip assembly forward, the gun mount will traverse clockwise. Check for erratic movement or surge while traversing in either direction. Increase gun mount speed to full slew and check maximum speed with stop watch.

4.5.1.31.4 Elevation and traverse system stability. To determine conformance to 3.5.12.1.4, with the rate motor toggle switch on drive controller turned to the ON position, the gun and mount shall be checked for movement in traverse and elevation with stop watch, azimuth indicator and gunners quadrant. The stability of gun and mount shall be as specified.

4.5.1.31.5 Manual elevation effort. To determine conformance to 3.5.12.1.5, attach a spring scale to hand crank; elevate and depress gun throughout full range of minus 5° to plus 87° and observe operation. Effort shall not exceed specified requirements.

4.5.1.31.6 Elevation mechanism backlash. To determine conformance to 3.5.12.1.6, the elevation mechanism shall be checked as follows:

a. If a backlash gage is available, place gun in a horizontal position and check backlash as described in (1) through (4) below:

   1. Install scale component of backlash gage on bearing adapter.
   2. Install vernier component of the backlash gage on output drive coupling and adjust knurled knob until gage reads zero.
   3. Twist the coupling rubber seal one way until the gun barrel begins to move; stop, and record the reading on the gage.

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4. Twist seal in opposite direction until slack is taken up and without moving the gun barrel; stop, and record reading on gage. A gunner's quadrant, boresight, or a suitable telescope clamped to the gun and sighted on a distant object should be utilized to detect motion of the barrel. Backlash shall be as specified.

b. If a backlash gage is not available, check backlash by first placing gun approximately horizontal. Mount a telescope temporarily on breech casing and proceed as described in (1) through (5) below:

1. Attach a wire to oil gear shaft to serve as a pointer.
2. Select a distant object; elevate gun and sight telescope so that the selected object appears at intersecting lines on the reticle. Come up to the distant object gradually and do not overpass.
3. Without moving the gun, scribe a line on the 5 3/8 inch diameter of the adapter housing to coincide with the wire pointer.
4. Elevate gun approximately one-half turn of elevation hand crank.
5. Depress gun slowly until the intersection of telescope reticle line falls on original reference point but do not overpass. Position of wire pointer must coincide with scribed line on housing within three-sixteenths of an inch.

4.5.1.31.7 Manual elevation limit stops. To determine conformance to 3.5.12.1.7, with the mount or carriage level the gun shall be operated to maximum depression and gun stop setting checked with gunners quadrant. The gun shall then be operated to maximum elevation and gun stop setting checked with gunners quadrant.

4.5.1.31.8 Power elevation. To determine conformance to 3.5.12.1.8, with mount or carriage level the guns shall be operated through the full range of travel and performance observed. The guns shall be brought smoothly to rest when slewed into either hydraulic elevation limit at a minimum speed of 25° per second and position of guns checked with gunners quadrant. Using stop watch operate guns in slew speed at specified positions and check for conformance to speed requirements.

CAUTION: Hydraulic stops shall be checked for proper setting before operating gun in full slew.

4.5.1.32 Fire control system test.

4.5.1.32.1 Synchronization. To determine conformance to 3.5.13.1, synchronization shall be checked as follows:

a. Place vehicle on as level terrain as possible and lock in azimuth.

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b. Prepare the gun mount for manual operation and make certain that the upper computer cover has been removed from the computing sight.
c. Release gun and gun mount locks.
d. Rotate the computer positioning handwheel of the computing sight until the plus 1600 mil graduation on the azimuth scale of the computer housing is alined with the azimuth scale index.
e. If the computing sight is a late (modified) model, set the speed knobs to the boresight mark ("B") at the 20 mph graduation on the knob scale and move the computer bail to 50° dive on the angle of flight scale. If the computing sight is an early (unmodified) model, set the speed knobs to the boresight mark at the 16 mph graduation on the knob scale and move the computer bail to 85° full dive on the angle of flight scale.
f. Place gunners quadrant M1 or M1A1 on the gun mount carriage leveling pads parallel to the dual gun with the line of fire arrow pointing toward the gun muzzle.
g. Adjust the gunners quadrant until the bubble in the level vial is centered. (This provides the basic setting for all the leveling procedures which follow.) Note the quadrant reading for future use.

NOTE: To prevent variations in reading during synchronization and/or boresighting, unnecessary movement of personnel on or over the vehicle, particularly the mount, is to be avoided.

h. Place the gunners quadrant on the breech casing leveling pads. Manually level the dual gun to the quadrant setting obtained in "g" above.

NOTE: Be certain to check the micrometer reading on the gunners quadrant to insure that the original setting was not disturbed.

i. Place gunners quadrant on computer body leveling pads and check level of computer body. This quadrant reading must be within 2 mils of quadrant reading obtained in "g" above. If reading is not within the prescribed limits, drag link rod must be adjusted.
j. The gunners quadrant shall read level within plus or minus 2 mils when placed on the mount carriage leveling pads, breech easing leveling pads, and the computer body leveling pads.
k. Manually elevate dual gun to 45° and check level of computer body. This reading must be level within plus or minus 2 mils.
l. Fully elevate dual gun and check level of computer body. From 45° to full elevation, this reading must be level within plus or minus 4 mils.

4.5.1.32.2 Boresighting check. To determine conformance to 3.5.13.2 and after dual gun has been synchronized, the accuracy of boresighting shall be checked as specified herein.

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a. With the upper computer cover removed, rotate the computer positioning handwheel to align the plus 1600 mil graduation on the azimuth scale of the computer housing with the azimuth scale index.
b. Set the speed knobs of modified computing sights to the boresight mark ("BS") at 20 mph graduation on the knob scale, and set the computer bail at 85° full dive.

NOTE: When setting the computer bail on a modified computing sight, move the bail to 60° dive, then back to 50° dive. This will eliminate variations in readings which result from inherent backlash in the sight.

c. Place gunners quadrant M1 or M1A1 on the gun mount carriage leveling pads parallel to the dual gun and with line of fire arrow pointing toward the gun muzzle.
d. Adjust the gunner's quadrant until the bubble in the level vial is centered. (This provides the basic setting for all the leveling procedure which follows.) Note the quadrant reading for future use.
e. Place the gunner's quadrant on the breech casing leveling pads. Manually level the dual gun to the quadrant setting obtained in "d" above.
f. Place the gunner's quadrant on the computer mechanism body leveling pads. If reading is not level within $\pm$ 1 mil, the vertical slide joint stem must be adjusted "g" and "h" below.
g. Loosen No. 5 x 1/2 roundhead screw which clamps computer cover to the base of the computer body and drop cover to expose end of vertical slide and slide joint stem. Loosen 1/4 x 3/4 cap screw in vertical slide and screw slide joint stem in or out until computer mechanism body is level within $\pm$ 1 mil.
h. Tighten the 1/4 x 3/4 cap screw in vertical slide. Position computer cover on computer body and secure by tightening the No. 5 x 1/2 round head screw.

NOTE: Excerise care that dust and dirt do not blow into the computer assembly while the computer cover is removed. After leveling the computer mechanism body, and before the cover is replaced, the accessible portion of the vertical slide and slide joint stem should be carefully wiped with a clean, lint-free cloth and lubricated with a light film of petroleum base hydraulic fluid (OHC), MIL-H-6083B, Type 1.

i. Set speed knob at zero mph and set computer bail at 0°. These settings will tilt the computer mechanism body forward 9 mils $\pm$ 1 mil, less than level. This operation sets in the proper superelevation.
j. Set the speed knobs to the 3S mark (20 mph on modified sights, 16 mph on unmodified sights) and the computer bail to 50° dive (modified sights) or 85° dive (unmodified sights).
k. With gunner's quadrant on the computer mechanism body leveling pads, check for level reading. If reading is not level within 1 mil, repeat "g" through "j" above.

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4.5.1.32.3 Boresighting procedure. To determine conformance to 3.5.13.2, the breech boresight M17-7692047 and muzzle boresight 5276694 shall be used for aligning the lines of sight of the sighting instruments with a line of sight through the axis of the gun bores upon a common aiming point no less than 1000 yards distance as follows:

a. Pull the hand operating lever of the left gun back as far as possible and fasten in this position while another man performs the operations outlined in "b" and "c" below.
b. *Open top cover of gun.*
c. Insert breech boresight M17 into chamber of left gun. Position boresight so the peep hole is up and press into place. Carefully move hand operating lever to SAFE position.

CAUTION: Exercise extreme care when installing breech boresight to prevent injury by accidently tripping the extractors.

d. Install muzzle boresight in muzzle of left gun with the crossed wires horizontal and vertical and the red disk hanging in view.
e. Set speed knobs to BS mark (20 mph) and computer bail to 50° dive. Set azimuth scale to plus 1600 mils.
f. Sighting through the boresights, move the dual gun until it is carefully aimed at a well defined object not less than 1000 yards distant.
g. Install reflex sight M24C.
h. Adjust reflex sight M24C so that its reticle pattern is accurately centered on the same distant aiming point (6) above.
i. Remove muzzle boresight. To remove breech boresight, pull hand operating lever out of SAFE position and back as far as it will move. Fasten lever back, reach into breech, and withdraw breech boresight. Move hand operating lever to firing position after breech has been completely cleared.
j. Close top cover to lift gun.

4.5.2 Control tests. Control tests for maintaining removal of those requirements whose inspection would be of too long duration to be included under acceptance tests shall be conducted by the manufacturer as specified herein (see 3.3.6).

4.5.2.1 Frequency. The first vehicle off the production line, or one of the first ten vehicles produced, and thereafter one per month or as specified, shall be checked in accordance with the following control tests.

4.5.2.2 Failure. If the vehicle selected fails to pass any of the control tests the Government inspector shall stop acceptance examination and testing on subsequent lots until the contractor furnishes evidence that an approved corrective action has been implemented.

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4.5.2.3 Contamination test. To determine conformance to 3.3.1.1, 3.3.1.2, 3.5.8, and 3.5.8.2, the vehicle after the water fording test, shall be examined for contamination of lubricants of the transmission, engine, final drive, and road wheels 2, 3, 4, 5 and one of their corresponding arms, on each side of the vehicle. Contamination of the lubricants shall not exceed 2 percent by volume.

4.5.2.4 Stowed equipment test. To determine conformance to 3.3.2, all manufacturer and depot installed OVE shall be stowed (combat loaded) on the selected vehicle. All items shall fit with proper clearance in the spaces provided. The equipment shall be removed after the test. The OVE, used shall be of the latest production available to the contractor; and the contractor shall be permitted to unpackage and repackage (for domestic shipment only) equipment used for test.

4.5.2.5 Fire extinguisher test. To determine conformance to 3.3.9, the fire extinguisher system shall be functionally tested. One vehicle selected from the first ten produced shall have the fire extinguisher tested both sides, subsequent tests of one in each fifty testing alternately the internal and external controls.

4.5.3 Inspection comparison tests (ICT). The Government may select vehicles any time during the contract production period and subject vehicles to tests listed on table IV and table V to reveal deficiencies of manufacture or design that may reduce the effective operation of these items in the field and to compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of vehicles shall be on a random spot check basis. Comparison test vehicles shall be combat loaded with all OVE stowed in their applicable spaces to conform to 3.3.2. Vehicles selected shall not include any vehicles previously tested for conformance to 4.5.2 (Control tests).

4.5.3.1 Test failure. Failure of any vehicle tested as specified in 4.5.3 to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occurring during, or as a result of the test, may be cause for refusal to continue acceptance of vehicles by the Government until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the conditions.

TABLE V. 2000 Mile test combat loaded.

Course	Mileage & speeds
Hard surface roads	300 miles at varying speeds up to maximum
Gravel and dirt roads	1300 miles at varying speeds up to maximum
Hilly cross country	200 miles at varying speeds up to maximum
Level cross country	200 miles at varying speeds up to maximum

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4.5.3.2 Fuel system test. To determine conformance to 3.3.4 and 3.3.4.1, the vehicle shall be operated throughout all speed and gear ranges over cross country, hilly terrain, and on smooth, level hard surface roads, and vehicle observed for functional requirements. Prior to performing this test, the fuel tank shall be filled, at a rate specified. The system shall be checked for leaks before and after a two mile road test. After road test the fuel tank shall be examined for foreign matter.

4.5.3.3 Exhaust system. To determine conformance to 3.3.5, the vehicle shall be "buttoned up" and the engine operated at 900 to 1000 rpm for a period of not less than 30 minutes. With the exhaust system in operation, the carbon monoxide (CO) content of the driver's compartment shall be analyzed.

4.5.3.4 Communications equipment test. To determine conformance to 3.3.13.2, communication system equipment shall be installed, subjected to tests and checked for voltage supply.

4.5.3.5 Sub-zero temperature tests. The vehicle, properly serviced and equipped, shall be subjected to applicable tests at the sub-zero temperatures specified in 3.5.1.

4.5.3.6 High temperature operation tests. The vehicle, properly serviced and equipped, shall be subjected to the applicable tests at the high temperatures specified in 3.3.3.1, and 3.5.1 and observed for performance and heating conditions, operating in ambient temperature of 115°F.

4.5.3.7 Speed on grades test. The vehicle shall be operated on grades of 20% and 3%. While ascending these grades it shall operate at the sustained speeds specified in 3.5.3.2.

4.5.3.8 Trench crossing tests. The vehicle selected shall be driven at a speed not to exceed 5 mph without stalling over trenches as specified in 3.5.9. The main gun shall be in stowed travel locked position, and in the forward fully elevated position. After the test the vehicle shall be examined for damage.

4.5.3.9 Vertical obstacle test. The vehicle selected shall be driven at a speed not to exceed 5 mph without stalling over vertical obstacles as specified in 3.5.10. The main gun shall be in stowed locked travel position, and in the forward fully elevated position. After the test the vehicle shall be examined for damage.

4.5.3.10 Climbing test. To determine conformance to 3.5.11, each vehicle selected shall be driven in forward and reverse gear up an approved 60% grade.

4.5.3.11 Radio suppression test. To determine conformance to 3.5.14, the vehicle shall be subjected to a radio interference suppression test (see 3.5.14).

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4.5.3.12 Carbon dioxide distribution check. To determine conformance to 3.3.9 and 4.5.2.4, the fire extinguisher system shall be tested and the results examined for adequate distribution and control. With the engine idling at 700 to 800 rpm, and CO_2 concentration meters located at the approximate center of each perforated distribution tube, the internal controls shall be operated as specified, using force gage. The first shot shall actuate one bottle of CO_2. After a time delay of not more than 40 seconds the system shall begin discharging CO_2 into the engine compartment, and the center of the Engine "V". After returning the handle to its original position, a second actuation shall discharge the remaining two bottles of CO_2. The CO_2 meters shall register a peak CO_2 concentration of at least 70% diminishing to zero within one minute after discharge. Removal and weighing of the CO_2 cylinders shall show a minimum of 11 pounds of CO_2 discharged.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. Vehicle shall be processed for storage and shipment in accordance with MIL-G-45178, or as specified by procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The vehicle is intended for use by the Armed Forces as a combat vehicle.

6.2 Ordering data. Contractual documents shall specify the following:

a. Title, number and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide fire-extinguisher cylinders. They should not be dropped, permitted to strike each other, or be handled roughly. Extreme care should be exercised during reinstallation to avoid tripping fire-extinguisher control system since physical injury is highly probable.

6.4 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is less than the AQL specified in the classification of defects for major and minor defects.

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of vehicles inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

6.5 Definitions. The following shall be used with the classification of defects for leaks:

1. Weep: Any evidence of fluid beyond the seal.
2. Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.

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3. Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet in the hydraulic system.
4. Droplets: Any evidence of fluid beyond the seal that results in the formation of 3 droplets in a 5 minute period.
5. Drip: Any evidence of fluid beyond the seal where droplet forms and falls.

6.6 Definitions of recurring major and minor defects. The Government Inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the Vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105.

Custodian:
Army - WC
Army - AT

Preparing activity:
Army - AT

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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